

best libraries in Python used for communication and interfacing the devices. For implementation using Python, the package 'paho-mqtt' is installed as follows:

```
$ pip install paho-mqtt
```

The following is the code snippet that can be integrated with IoT based devices and gadgets:

```
import paho.mqtt.client as mqtt
# Define event callbacks
def on_connect(client, mySignalData,
CN):
    if CN == 0:
        print("Connected
successfully.")
    else:
        print("Connection failed. CN=
"+str(CN))
def on_publish(client, mySignalData,
MyString):
    print("Message "+str(MyString)+"
published.")
def on_subscribe(client, mySignalData,
MyString, granted_qos):
    print("Subscribe with MyString
"+str(MyString)+" received.")
def on_message(client, mySignalData,
msg):
    print("Message received on topic
"+msg.topic+" with QoS "+str(msg.qos)+"
and payload "+msg.payload)
mqttclient = myMQTT.Client()
# Assign event callbacks
mqttclient.on_connect = on_connect
mqttclient.on_publish = on_publish
mqttclient.on_subscribe = on_subscribe
mqttclient.on_message = on_message
# Connect
mqttclient.username_pw_
set('kumargaurav.in@gmail.com',
'7881730a')
mqttclient.connect('myMQTT.dioty.co',
1883)
# Subscribing
mqttclient.subscribe('/kumargaurav.in@
gmail.com/')
# Publishing a message
x=input('Message')
```

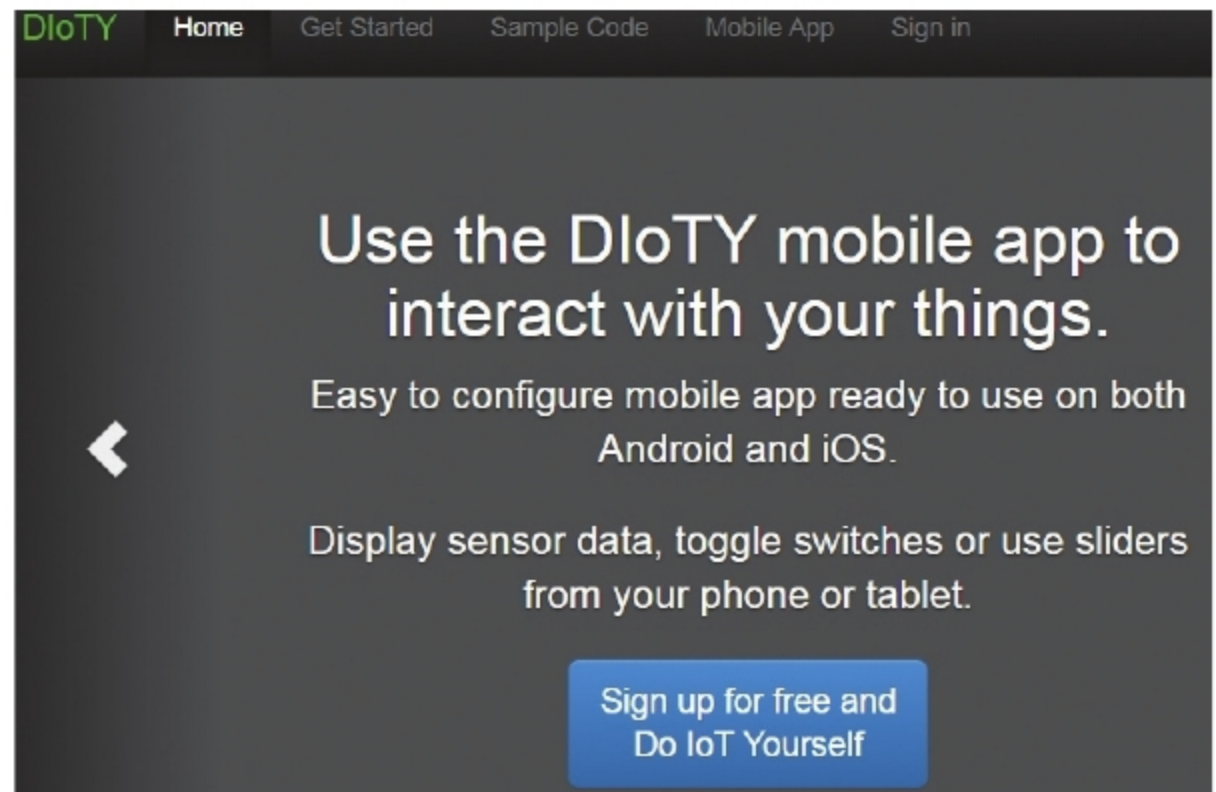


Figure 5: Dashboard of DIoT

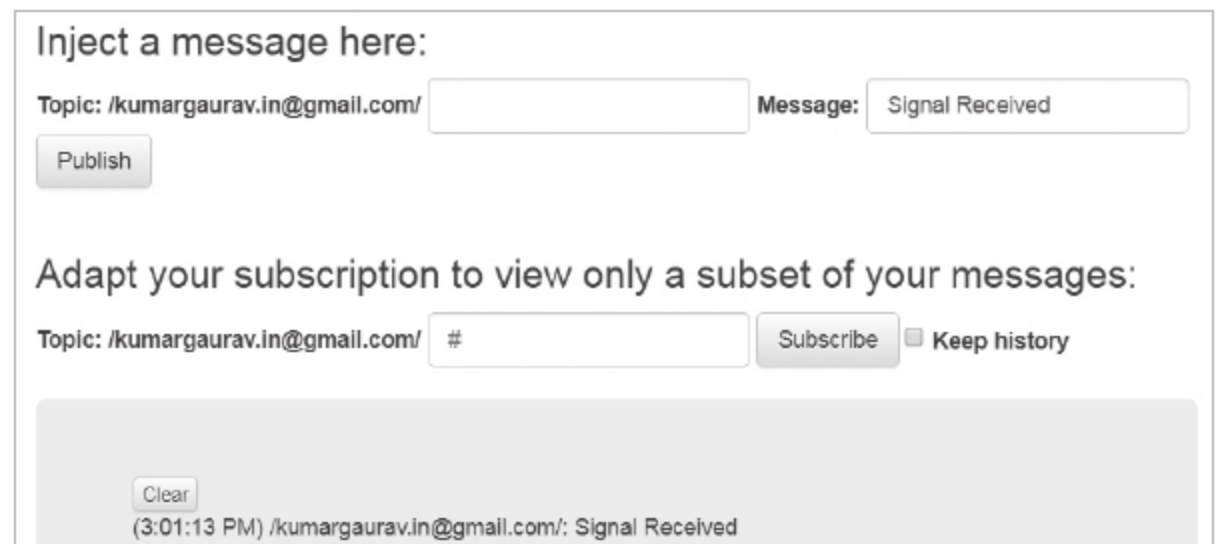


Figure 6: Receiving data on the DIoT dashboard using MQTT

```
mqttclient.publish('/kumargaurav.in@
gmail.com/', x)
# Loop; exit on error
CN = 0
while CN == 0:
    CN = mqttclient.loop()
    print("CN: " + str(CN))
```

After executing the Python script, the data is handled and processed by MQTT broker, as shown in Figure 6.

Scope for research and development

Research scholars, academicians and practitioners can carry out research and make innovative problem statements on IoT using MQTT brokers. A number of research statements and domains exist in which MQTT based implementations can be done including telemedicine, personal security gadgets, defence equipment, military devices, agribots for smart agriculture, and many others. **END** 🐧

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